

EDA-schema V2:

A Multimodal Schema, Open Datasets, and Benchmarks for Machine Learning in Digital Physical

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EDA-Schema V2

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- Arxiv Paper

- <https://arxiv.org/pdf/2605.06952>

- Github Repository

- <https://github.com/drexel-ice/eda-schema>

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EDA-Schema-V2: A Multimodal Schema, Open Datasets, and Benchmarks for Machine Learning in Digital Physical Design

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The continuous scaling of CMOS technology has significantly increased the complexity of very large-scale integrated (VLSI) circuits, driving growing interest in applying machine learning (ML) algorithms to electronic design automation (EDA). However, the limited availability of open datasets and the absence of standardized data representations present major challenges to the research community. In particular, the lack of interoperability and comparability among ML-based research in digital physical design hinders reproducibility and collaborative advancement. This paper introduces *EDA-Schema-V2*, an open and comprehensive multimodal graph and image-based schema designed to address interoperability and comparability by providing a structured framework for the representation and analysis of datasets in digital physical design. The schema includes representations of both physical attributes and quality-of-results (QoR) metrics of circuits across multiple stages of the design flow, including logic synthesis, floorplanning, placement, clock network synthesis, and routing.

README Code of conduct Contributing License Security

EDA-Schema: A Multimodal Datamodel for Digital Circuit Design

build: passing python: 3.11+ license: CC BY-NC-SA 4.0

EDA-Schema is an open, standardized, and extensible **multimodal datamodel and framework** for representing, storing, and analyzing digital circuit physical design data across the RTL to GDSII flow. It models circuits using **heterogeneous graphs**, **spatial image modalities**, **scalar heatmaps**, and **structured metric entities**, providing unified representations of circuit structure, electrical behavior, quality-of-results metrics, and stage-wise design evolution for reproducible analysis and machine learning research in electronic design automation.

Key Features

- Unified Multimodal Datamodel



EDA-Schema V2: What the paper + Github repository contributes?

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1. Standardized Datamodel Schema

- Unified representation of design flow, circuit structure, and QoR data
- Converts raw EDA outputs into ML-ready graph, image, and tabular formats

2. Open Multimodal Dataset

- Built with OpenROAD across four open-source PDKs (Nangate 45 nm, SkyWater 130 nm, IHP SG13G2 130 nm, and ASAP 7 nm)
- Includes IWLS'05 designs with systematic physical design parameter sweeps

3. Baseline ML Benchmarks

- Defines prediction tasks for timing, power, area, and routing metrics
- Provides baseline results for reproducible model comparison

4. Python Library and Resources

- Provides Python APIs to load EDA-Schema datasets, access schema entities, and query graph, image, and tabular data
- Includes examples, tutorials, and notebooks for data exploration and visualization

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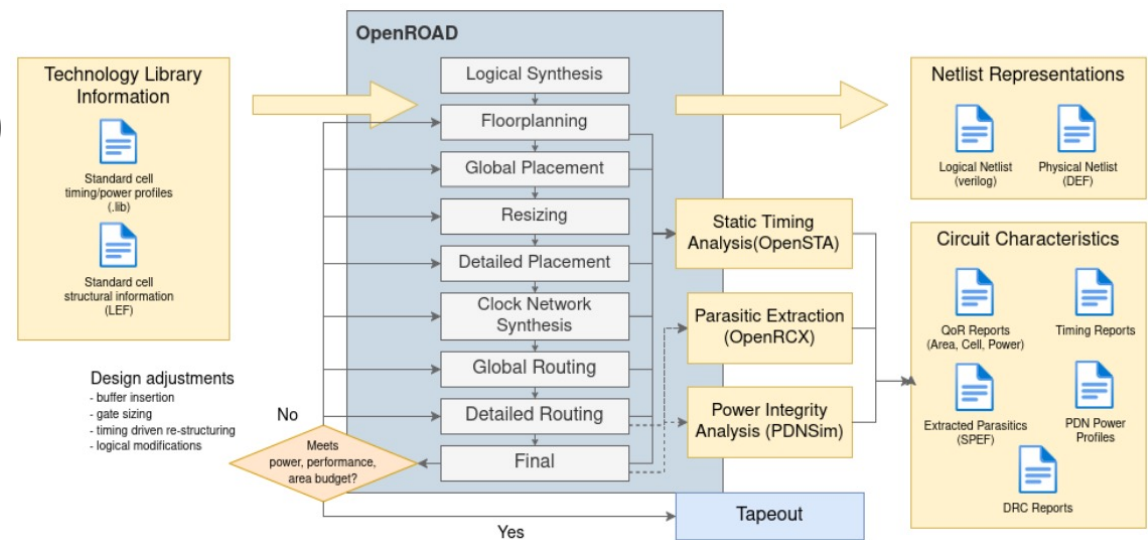
RTL-to-GDSII data pipeline

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- RTL-to-GDSII design flows generate artifacts needed for schema and benchmark labels
- EDA-Schema organizes the artifacts into standardized schema entities and labels
- In this work, OpenROAD is used to generate the dataset artifacts

Key flow outputs

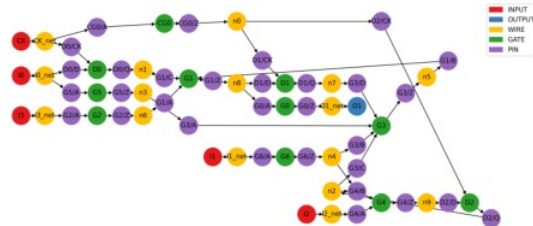
- Logical netlist (Verilog/Liberty cell data)
- Physical netlist (LEF and DEF files)
- QoR reports
- Timing analysis (OpenSTA)
- Parasitics analysis (OpenRCX)
- EM/IR analysis (PDNSim)



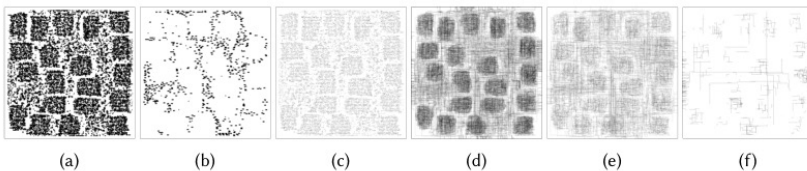
Multimodal representation of Netlist

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- Netlist graph
 - Nodes represent gates, nets, pins and I/O ports
 - Gate attributes (cell type, geometry, placement ...)
 - Net attributes (fanout, hpwl, physical routing, RC ...)

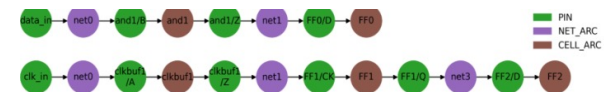


Sample circuit as a netlist graph



Binary spatial maps for cell placement, pins, and routing

- STA-derived path graph
 - Models each timing path as a directed graph from startpoint to endpoint
 - Nodes represent pins, I/O ports, cell timing arcs, and net timing arcs
 - Cell arcs capture gate delay, slew, and arrival time
 - Net arcs capture RC interconnect delay, slew, capacitance, and arrival time



Example timing path graph with pin, net arc, and cell arc entities

9:33 AM

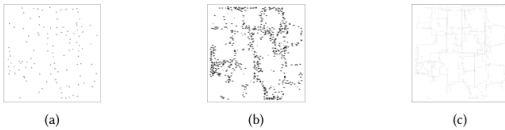
Other Primary Representations

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- EDA-Schema-V2 keeps analysis maps aligned with physical layout using technology-aware resolution rules

Clock network

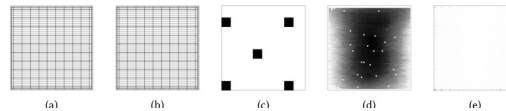
Captures clock source-to-sink structure, clock buffer placement, sequential elements, and clock routing.



Clock buffers, sequential cells, and clock routes

Power delivery

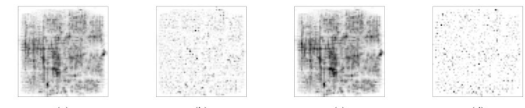
Models power grid geometry and analysis results from PDNSim, including IR-drop and electromigration.



VDD/VSS routing, sources, IR-drop, and EM

Routability

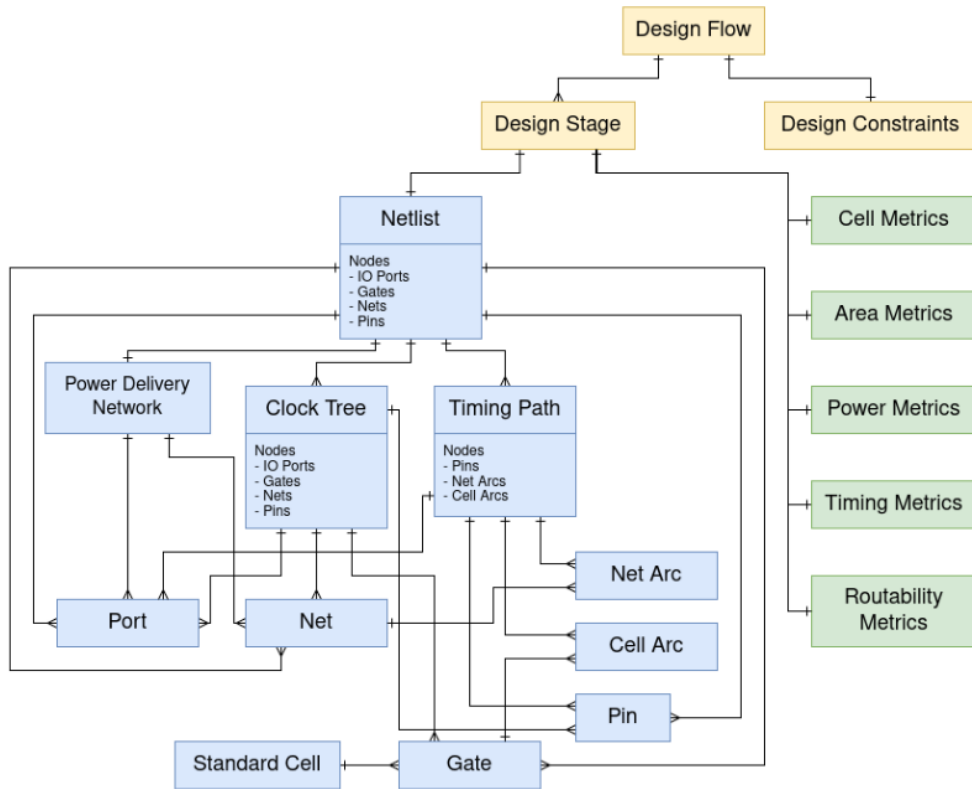
Stores scalar RUDY maps that estimate routing demand and congestion across tiles.



RUDY, pin RUDY, long-range, and short-range demand

Entity Relationship Diagram

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Entity	Feature	Datatype	Unit	Stages
Design Flow	id	string		
	taskname	string		
	design	string		
	run_status	string		
Constraints	clock_period	float	ns	
	clock_uncertainty	float	ns	
	clock_latency	float	ns	
	clock_transition	float	ns	
	input_delay	float	ns	
	output_delay	float	ns	
	aspect_ratio	float		
	utilization	float		
	run_status	string		
	direction	string		
Design Stage	width	float	µm	PP-F
	height	float	µm	PP-F
	no. of inputs	int		PP-F
	no. of outputs	int		PP-F
	no. of cells	int		PP-F
	no. of nets	int		PP-F
	no. of pins	int		PP-F
	utilization	float		PP-F
	total_wirelength	float	µm	DR-F
	total_area	float	µm	PP-F
Netlist	cell_placement	binary map	CP-F	
	cell_placement_combinational	binary map	CP-F	
	cell_placement_sequential	binary map	CP-F	
	cell_placement_filler	binary map	F	
	pin_placement	binary map	CP-F	
	routing	binary map	DR-F	
	routing_by_metal_layers	int(binary map)	DR-F	
	clock_source	string	PP-F	
	no. of buffers	int	CIS-F	
	no. of clock sinks	int	CIS-F	
Clock Tree	cell_placement	binary map	CIS-F	
	cell_placement_combinational	binary map	CIS-F	
	cell_placement_sequential	binary map	CIS-F	
	pin_placement	binary map	CIS-F	
	routing	binary map	DR-F	
	routing_by_metal	int(binary map)	DR-F	
	pdn_routing_vdd	binary map	PP-F	
	pdn_routing_vss	binary map	PP-F	
	voltage_source	binary map	PP-F	
	tr_drop_vdd	scalar map	mV	DR-F
Power Delivery Network	tr_drop_vss	scalar map	mV	DR-F
	em_vdd	scalar map	mV	DR-F
	em_vss	scalar map	mV	DR-F
	name	string		
	direction	string		
	x	float	µm	CP-F
	y	float	µm	CP-F
	functions	string		
	width	float	µm	
	height	float	µm	
Port	no. of input pins	int		
	is_sequential	bool		
	is_inverter	bool		
	is_buffer	bool		
	is_filler	bool		
	is_diode	bool		
	drive_strength	float		
	input_capacitance_min	float	fF	
	input_capacitance_max	float	fF	
	output_capacitance_min	float	fF	
Standard Cell	output_capacitance_max	float	fF	
	leakage_power_min	float	µW	
	leakage_power_max	float	µW	
	name	string		
	standard_cell	Standard cell	PP-F	
	x_min	float	µm	CP-F
	y_min	float	µm	CP-F
	x_max	float	µm	CP-F
	y_max	float	µm	CP-F
	no. of inputs	int		PP-F
Gate	no. of outputs	int		PP-F
	internal_power	float	µW	PP-F
	switching_power	float	µW	PP-F
	leakage_power	float	µW	PP-F
	total_power	float	µW	PP-F
	tr_drop_vdd	float	mV	DR-F
	tr_drop_vss	float	mV	DR-F
	no. of endpoints	int		PP-F
	no. of violating endpoints	int		PP-F
	nudy_net	scalar map	DR-F	

Entity	Feature	Datatype	Unit	Stages
Net	name	string		PP-F
	is_special_net	bool		PP-F
	no. of inputs	int		PP-F
	x_min	float	µm	CP-F
	y_min	float	µm	CP-F
	x_max	float	µm	CP-F
	y_max	float	µm	CP-F
	length	float	µm	DR-F
	hpol	float	µm	CP-F
	resistance	float	ohm	DR-F
Pin	capacitance	float	fF	DR-F
	total_coupling_capacitance	float	fF	DR-F
	name	string		PP-F
	direction	string		PP-F
	x_min	float	µm	CP-F
	y_min	float	µm	CP-F
	x_max	float	µm	CP-F
	y_max	float	µm	CP-F
	is_startpoint	bool		PP-F
	is_endpoint	bool		PP-F
Timing Path	setup_rise_slew	float	ns	PP-F
	setup_fall_slew	float	ns	PP-F
	hold_rise_slew	float	ns	PP-F
	hold_fall_slew	float	ns	PP-F
	setup_rise_slack	float	ns	PP-F
	setup_fall_slack	float	ns	PP-F
	hold_rise_slack	float	ns	PP-F
	hold_fall_slack	float	ns	PP-F
	load_capacitance	float	fF	PP-F
	switching_activity	float		PP-F
Cell Arc	startpoint	string		PP-F
	endpoint	string		PP-F
	path_type	string		PP-F
	arrival_time	float	ns	PP-F
	required_time	float	ns	PP-F
	slack	float	ns	PP-F
	no. of pins	int		PP-F
	is_critical_path	bool		PP-F
	gate	Gate Instance		PP-F
	delay	float	ns	PP-F
Net Arc	arrival_time	float	ns	PP-F
	slew	float	ns	PP-F
	net	Net		PP-F
	delay	float	ns	PP-F
	arrival_time	float	ns	PP-F
	slew	float	ns	PP-F
	capacitance	float	fF	PP-F
	no. of combinational cells	int		PP-F
	no. of sequential cells	int		PP-F
	no. of buffers	int		PP-F
Cell Metrics	no. of inverters	int		PP-F
	no. of fillers	int		PP-F
	no. of top cells	int		PP-F
	no. of diodes	int		PP-F
	no. of macros	int		PP-F
	no. of total cells	int		PP-F
	combinational_cell_area	float	µm ²	PP-F
	sequential_cell_area	float	µm ²	PP-F
	buffer_area	float	µm ²	PP-F
	inverter_area	float	µm ²	PP-F
Area Metrics	filler_area	float	µm ²	PP-F
	top_cell_area	float	µm ²	PP-F
	diode_area	float	µm ²	PP-F
	macro_area	float	µm ²	PP-F
	cell_area	float	µm ²	PP-F
	total_area	float	µm ²	PP-F
	combinational_power	float	µW	PP-F
	sequential_power	float	µW	PP-F
	macro_power	float	µW	PP-F
	internal_power	float	µW	PP-F
Power Metrics	switching_power	float	µW	PP-F
	leakage_power	float	µW	PP-F
	total_power	float	µW	PP-F
	total_negative_slack	float	ns	PP-F
	worst_slack	float	ns	PP-F
	worst_arrival_time	float	ns	PP-F
	worst_required_time	float	ns	PP-F
	critical_path_startpoint	string		PP-F
	critical_path_endpoint	string		PP-F
	no. of endpoints	int		PP-F
Timing Metrics	no. of violating endpoints	int		PP-F
	nudy_net	scalar map	DR-F	
	nudy_net_long	scalar map	DR-F	
Routability Metric	nudy_net_short	scalar map	DR-F	
	nudy_pin	scalar map	DR-F	

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Open Dataset

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- Setup
 - Designs: 18 IWLS'05 benchmark circuits
 - PDK:
 - SkyWater 130 nm
 - Nangate 45 nm
 - IHP SG13G2 130 nm
 - ASAP 7 nm
 - Design Toolset: OpenROAD
 - Parameter sweep: 108 parameter setups per design
- Data generated is parsed and mapped into EDA-schema as python objects

IWLS'05 benchmark circuit characteristics

Circuits	No. of inputs	No. of outputs	No. of Registers
ac97_ctrl	84	48	2211
aes_core	259	129	562
des3_area	240	64	64
ethernet	96	115	10544
i2c	19	14	129
jpeg	20	27	4383
mem_ctrl	115	152	1083
pci	162	207	3220
sasc	16	12	118
simple_spi	16	12	131
spi	47	45	229
ss_pcm	19	9	87
systemcaes	260	129	670
systemcdes	132	65	190
tv80	14	32	361
usb_funct	128	121	1740
usb_phy	15	18	108
wb_dma	217	215	521

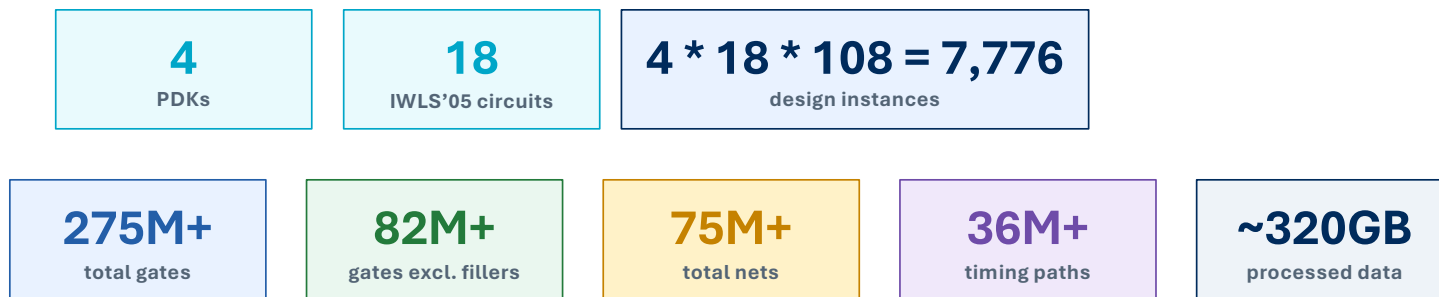
Dataset constraints and parameters

Parameters	Values or Ranges			
Target clock periods (ns)	{0.8 × BF, BF, BP, 1.2 × BP}			
Aspect ratio	{0.5, 1, 1.5}			
Core Utilization	NG45	SKY130	IHP130	ASAP7
	{0.3, 0.4, 0.5}	{0.2, 0.3, 0.4}	{0.2, 0.3, 0.4}	{0.3, 0.4, 0.5}
Core Margin (μm)	NG45	SKY130	IHP130	ASAP7
	5.72	22.4	22.72	2.29
Placement Density	Uniform, 1.25 × Uniform, 1.5 × Uniform			
Input/Output Delay (ns)	20% of Target clock period (TCP)			
Clock Latency (ns)	1% of TCP (capped at 50 ps)			
Clock Transition (ns) (12.5% of TCP of the fastest circuit for a PDK)	NG45	SKY130	IHP130	ASAP7
	0.05	0.1875	0.1875	0.02375
Clock Uncertainty (ns)	5% of TCP (capped at 50 ps)			

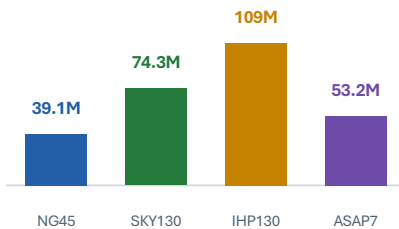
Dataset at a glance

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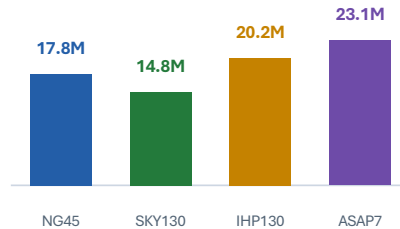
- [Dataset link](#)



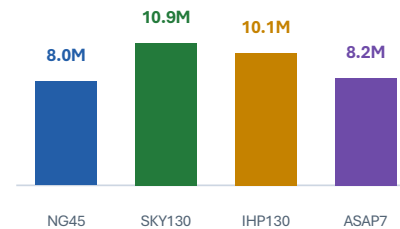
Total gates by PDK, millions



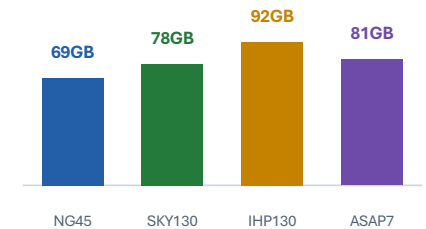
Total nets by PDK, millions



Timing paths by PDK, millions



Processed dataset size by PDK



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Benchmark Question: Can early-stage data predict final design quality?

- Intermediate-stage estimates from OpenROAD are compared against detailed-route values.
- Future ML models are expected to outperform these baselines.

Circuit-level QoR

- Total area
- Total power
- Total wirelength

Net-level routing

- Interconnect wirelength

Design-level timing

- Worst arrival time
- Worst slack
- Total negative slack

Path-level timing

- Timing path arrival time
- Timing path slack

Arc-level timing

- Net arc delay
- Cell arc delay
- Cell arc slew

EDA-Schema V2 Baselines

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- Baseline evaluates 12 early-stage metric prediction tasks based on the toolset results (OpenROAD)

- Any ML model should beat these numbers to be considered useful

- Seed leaderboard

	Metrics	floorplan to detailed route				global place to detailed route				detailed place to detailed route				CTS to detailed route				global route to detailed route			
		NG45	SKY130	IPH130	ASAP7	NG45	SKY130	IPH130	ASAP7	NG45	SKY130	IPH130	ASAP7	NG45	SKY130	IPH130	ASAP7	NG45	SKY130	IPH130	ASAP7
Total Area (μm ²)	MAE	1.781.97	18,567.03	48,738.62	225.06	871.07	8,385.61	35,724.45	73.79	571.13	7,505.03	19,487.21	47.23	49.02	854.81	2,339.67	1.98	0.00	0.00	0.00	0.00
	MAPE	12.43 %	22.51 %	20.74 %	15.35 %	7.07 %	11.71 %	13.60 %	7.50 %	4.14 %	9.27 %	5.56 %	3.79 %	0.49 %	1.22 %	0.72 %	0.24 %	0.00 %	0.00 %	0.00 %	0.00 %
	R ²	0.989	0.957	0.912	0.969	0.997	0.991	0.945	0.997	0.998	0.991	0.979	0.998	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	R ²	11,875.86	46,947.57	17,699.64	6,071.61	5,670.15	19,274.83	16,651.34	2,266.72	5,043.04	19,382.95	15,644.47	2,177.82	280.22	2,270.66	123.65	55.15	0.00	0.00	0.00	0.00
Total Power (μW)	MAE	27.89 %	50.85 %	23.71 %	33.47 %	16.70 %	31.36 %	18.87 %	19.23 %	15.37 %	30.00 %	16.83 %	17.57 %	0.57 %	1.67 %	0.24 %	0.29 %	0.00 %	0.00 %	0.00 %	0.00 %
	MAPE	0.948	0.648	0.807	0.823	0.987	0.946	0.781	0.976	0.989	0.944	0.798	0.977	1.000	0.997	1.000	1.000	1.000	1.000	1.000	1.000
	R ²	30,529.92	83,777.43	117,233.52	6,460.04	20,115.77	61,602.30	57,977.64	3,479.88	13,920.52	40,751.29	26,046.32	2,562.35	13,698.67	38,941.02	25,473.38	2,491.87				
	R ²	29.21 %	34.24 %	24.43 %	17.17 %	19.61 %	24.30 %	11.49 %	7.05 %	13.16 %	16.18 %	5.51 %	3.40 %	12.53 %	14.86 %	5.33 %	3.28 %				
Total wirelength (μm)	MAE	0.916	0.902	0.928	0.987	0.959	0.939	0.977	0.995	0.980	0.975	0.996	0.997	0.981	0.977	0.996	0.997				
	MAPE	4.83	12.41	13.19	1.45	4.13	11.8	11.09	1.41	4.19	12.01	11.25	1.44	4.19	11.84	11.17	1.43				
	R ²	87.59 %	94.04 %	72.60 %	113.14 %	81.33 %	86.04 %	73.29 %	125.55 %	78.68 %	72.85 %	71.34 %	123.19 %	78.20 %	69.82 %	70.16 %	122.69 %				
	R ²	0.735	0.751	0.89	0.886	0.77	0.773	0.916	0.892	0.777	0.791	0.915	0.893	0.779	0.798	0.918	0.895				
Interconnect length (μm)	MAE P95	12.72	38.53	53.88	5.57	10.32	39.23	46.46	5.44	10.82	41.92	46.28	5.85	10.79	41.09	46.08	5.83				
	MAPE P95	230.82 %	293.66 %	339.20 %	410.10 %	186.85 %	247.83 %	326.04 %	403.03 %	181.48 %	238.43 %	291.33 %	402.78 %	180.77 %	234.56 %	291.33 %	402.78 %				
	MAE TOP5	44.73	116.02	86.57	11.67	46.91	117.87	79.59	11.37	47.4	121.41	80.45	11.65	47.61	121.04	80.05	11.61				
	MAPE TOP5	24.45 %	25.39 %	19.20 %	18.59 %	23.99 %	24.99 %	18.51 %	18.19 %	24.18 %	25.04 %	18.56 %	19.03 %	24.12 %	24.74 %	18.49 %	18.90 %				
Worst Arrival Time (ns)	MAE	3.10	10.84	8.55	0.09	0.16	0.77	0.41	0.06	0.13	1.21	0.34	0.06	0.01	0.14	0.06	0.01	0.00	0.00	0.00	0.00
	MAPE	192.92 %	183.71 %	331.31 %	15.78 %	16.19 %	21.48 %	16.98 %	11.17 %	13.81 %	33.30 %	15.82 %	10.44 %	1.60 %	4.69 %	2.32 %	2.14 %	0.00 %	0.00 %	0.00 %	0.00 %
	R ²	3.18	11.28	8.83	0.03	0.21	1.15	0.27	0.05	0.15	1.62	0.31	0.05	0.01	0.09	0.04	0.01	0.00	0.00	0.00	0.00
	R ²	0.06	0.30	0.30	0.04	0.09	0.27	0.27	0.04	0.04	0.17	0.20	0.03	0.01	0.03	0.04	0.00	No positive or negative error			
Worst Slack (ns)	MPE	3.68	12.15	9.35	0.02	0.24	1.26	0.26	0.05	0.17	1.68	0.36	0.06	0.02	0.12	0.13	0.01	$n_p = n_n = 0$			
	TPR	95.67 %	96.32 %	94.52 %	67.94 %	95.67 %	96.21 %	64.38 %	93.08 %	95.67 %	98.53 %	58.90 %	99.45 %	100.00 %	99.47 %	96.58 %	100.00 %	100.00 %	100.00 %	100.00 %	100.00 %
	TNR	28.70 %	15.76 %	28.11 %	97.70 %	57.56 %	21.39 %	95.54 %	63.75 %	56.26 %	8.39 %	91.58 %	58.79 %	93.24 %	60.59 %	99.22 %	91.53 %	100.00 %	100.00 %	100.00 %	100.00 %
	TNR	1.764.18	17,768.52	3,755.75	1.09	57.74	727.04	0.57	5.65	22.45	959.12	1.05	5.75	2.11	70.22	0.04	1.01	0.00	0.00	0.00	0.00
Total Negative Slack (ns)	MPE	1.26	4.74	2.50	3.92	24.60	5.30	1.82	3.76	23.22	2.74	1.88	4.68	4.81	22.51	0.17	1.56	No positive or negative error			
	MNE	2,296.57	20,176.54	5,201.66	2.10	101.63	849.90	7.15	11.49	37.22	1,031.87	9.00	10.54	6.34	123.39	0.76	3.78	$n_p = n_n = 0$			
	MAE	0.4756	2.9814	1.7195	0.1836	0.2313	0.9552	0.6682	0.1509	0.2398	0.9151	0.6392	0.1462	0.0022	0.0319	0.0041	0.0012	0.0000	0.0000	0.0000	0.0000
	MAPE	75.89 %	92.31 %	95.11 %	51.92 %	46.25 %	39.61 %	43.00 %	42.55 %	45.63 %	37.87 %	41.34 %	41.05 %	0.27 %	0.80 %	0.18 %	0.22 %	0.00 %	0.00 %	0.00 %	0.00 %
Timing Path Arrival Time (ns)	MAE P95	1.0541	7.1590	4.6540	0.3296	0.3600	1.5150	0.8949	0.2385	0.3570	1.5320	0.8899	0.2387	0.0107	0.1980	0.0220	0.0036	0.0000	0.0000	0.0000	0.0000
	MAPE P95	133.31 %	169.45 %	241.72 %	74.00 %	69.54 %	64.36 %	65.89 %	70.09 %	69.32 %	64.39 %	65.77 %	70.06 %	1.27 %	4.81 %	0.98 %	0.68 %	0.00 %	0.00 %	0.00 %	0.00 %
	MAE TOP5	2.6279	10.1956	1.8427	0.3048	0.2398	0.9467	0.7733	0.1986	0.1971	1.0037	0.7550	0.1975	0.0107	0.1110	0.0246	0.0037	0.0000	0.0000	0.0000	0.0000
	MAPE TOP5	208.58 %	164.89 %	58.23 %	36.41 %	15.32 %	14.73 %	24.60 %	23.24 %	12.57 %	15.69 %	23.97 %	23.01 %	0.70 %	1.68 %	0.73 %	0.42 %	0.00 %	0.00 %	0.00 %	0.00 %
Timing Path Slack (ns)	MAE	0.2450	2.6909	1.8879	0.0400	0.0532	0.2179	0.0736	0.0166	0.0280	0.2873	0.0975	0.0203	0.0023	0.0316	0.0040	0.0012	0.0000	0.0000	0.0000	0.0000
	MPE	0.0361	1.7111	0.9992	0.0788	0.0282	0.0851	0.0556	0.0150	0.0299	0.1210	0.0741	0.0167	0.0125	0.0351	0.0159	0.0080	No positive or negative error			
	MNE	0.2749	2.8765	2.8290	0.0222	0.0641	0.4140	0.1082	0.0203	0.0273	0.5783	0.1349	0.0296	0.0217	0.1342	0.0167	0.0149	$n_p = n_n = 0$			
	TPR	98.01 %	96.65 %	74.89 %	70.21 %	96.50 %	98.01 %	74.19 %	83.93 %	94.90 %	97.61 %	73.80 %	85.13 %	93.04 %	95.99 %	84.56 %	95.92 %	100.00 %	100.00 %	100.00 %	100.00 %
Net Arc Delay (ns)	TNR	88.69 %	77.61 %	93.99 %	99.96 %	96.00 %	90.69 %	99.93 %	98.69 %	97.65 %	98.43 %	99.92 %	98.56 %	99.63 %	98.34 %	99.99 %	99.73 %	100.00 %	100.00 %	100.00 %	100.00 %
	MAE	0.1128	0.7713	0.4259	0.0105	0.0200	0.0631	0.0306	0.0066	0.0122	0.0466	0.0165	0.0039	0.0005	0.0046	0.0004	0.0002	0.0000	0.0000	0.0000	0.0000
	MAPE	>10000 %	403.06 %	547.23 %	39.38 %	>10000 %	28.21 %	31.14 %	20.69 %	>10000 %	19.99 %	8.91 %	11.16 %	>10000 %	2.36 %	0.32 %	0.95 %	0.00 %	0.00 %	0.00 %	0.00 %
	R ²	<-1	<-1	<-1	<-1	<-1	-0.029	<-1	-0.402	-0.024	0.748	0.876	0.777	0.976	0.981	0.999	0.979	1.000	1.000	1.000	1.000
Cell Arc Delay (ns)	MAE	0.0011	0.0031	0.0003	0.0018	0.0008	0.0016	0.0002	0.0003	0.0009	0.0013	0.0002	0.0001	0.0003	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	MAPE	>10000 %	>10000 %	>10000 %	>10000 %	>10000 %	>10000 %	>10000 %	>10000 %	>10000 %	>10000 %	>10000 %	>10000 %	>10000 %	>10000 %	>10000 %	>10000 %	0.00 %	0.00 %	0.00 %	0.00 %
	R ²	-0.308	<-1	<-1	-0.141	0.928	0.873	0.641	0.851	0.875	0.894	0.657	0.925	0.987	0.974	0.975	0.952	1.000	1.000	1.000	1.000
	R ²	0.0761	0.8919	0.4735	0.0152	0.0082	0.0415	0.0059	0.0026	0.0028	0.0392	0.0011	0.0017	0.0003	0.0054	0.0004	0.0003	0.0000	0.0000	0.0000	0.0000
Cell Arc Slew (ns)	MAE	183.03 %	462.31 %	1134.69 %	34.33 %	27.44 %	37.69 %	11.05 %	7.44 %	13.43 %	35.60 %	2.05 %	5.89 %	1.27 %	4.66 %	0.53 %	1.01 %	0.00 %	0.00 %	0.00 %	0.00 %
	R ²	<-1	<-1	<-1	<-1	0.332	0.414	0.965	0.878	0.786	0.427	0.987	0.911	0.973	0.938	0.994	0.964	1.000	1.000	1.000	1.000

EDA-Schema V2: What the paper + Github repository contributes?

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1. Standardized Datamodel Schema

- Unified representation of design flow, circuit structure, and QoR data
- Converts raw EDA outputs into ML-ready graph, image, and tabular formats

2. Open Multimodal Dataset

- Built with OpenROAD across four open-source PDKs (Nangate 45 nm, SkyWater 130 nm, IHP SG13G2 130 nm, and ASAP 7 nm)
- Includes IWLS'05 designs with systematic physical design parameter sweeps

3. Baseline ML Benchmarks

- Defines prediction tasks for timing, power, area, and routing metrics
- Provides baseline results for reproducible model comparison

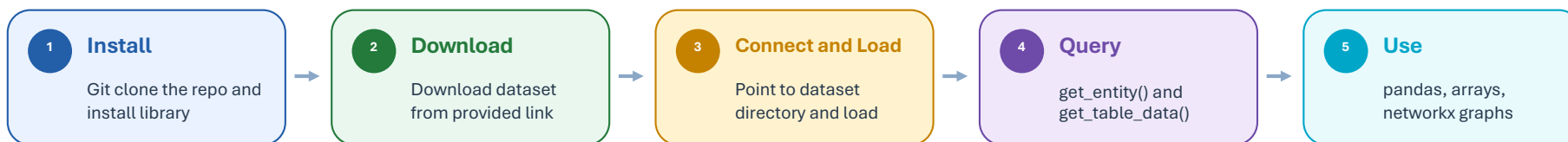
4. Python Library and Resources

- Provides Python APIs to load EDA-Schema datasets, access schema entities, and query graph, image, and tabular data
- Includes examples, tutorials, and notebooks for data exploration and visualization

EDA-Schema Python Repository

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- Github Repo: <https://github.com/drexel-ice/EDA-schema>



Minimal code path

```
from eda_schema.dataset import Dataset
from eda_schema.db.parquet import ParquetDB

db = ParquetDB("dataset/test")
dataset = Dataset(db)
dataset.load()

print(f"Available flows: {len(dataset)}")
```

Resources

- [examples/](#)
Start here for loading entities and basic query patterns
- [research/notebooks/tutorials/](#)
Notebook workflows for data analysis, visualization, and exploration
- [research/eda-schema-v2/](#)
resource to reproduce paper results

Querying Schema Entities

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- One API surface for graph entities, QoR tables, and spatial arrays

Entity access

```
netlist =  
dataset.db.get_entity(  
    "netlists",  
    flow_id="gcd-000001",  
    stage="cts",  
)  
  
print(netlist.no_of_cells)  
print(netlist.no_of_nets)  
print(netlist.no_of_pins)
```

Graph modality

```
netlist = dataset.db.get_entity(  
    "netlists",  
    flow_id="gcd-00001",  
    stage="cts",  
)  
  
gates = [n for n, d in  
    netlist.nodes(data=True)  
    if d["type"] == "GATE"]  
  
nets = [n for n, d in  
    netlist.nodes(data=True)  
    if d["type"] == "NET"]  
  
edges = list(netlist.edges)  
subgraph =  
    netlist.subgraph(gates[:10])
```

Spatial modalities

```
netlist =  
dataset.db.get_entity(  
    "netlists",  
    flow_id="gcd-000001",  
    stage="global_route",  
)  
  
cell_map =  
    netlist.cell_placement  
print(cell_map.shape)  
cell_map.plot()
```

Tabular/Metric access

```
timing =  
dataset.db.get_table_data(  
    "timing_metrics",  
    flow_id="gcd-000001",  
    stage="detailed_route",  
)  
  
print(timing.worst_slack.min()  
)  
print(timing.total_negative_s  
    lack.sum())
```

What users get back

Python objects

Netlist, Clock Network, Timing
Path, PDN, metric entities

Networkx Graphs

Netlist, clock network, and timing
path graph structures

2D Numpy Arrays

Placement, pin, routing, PDN, IR-
drop, EM, and RUDY maps

DataFrames

Stage-filtered QoR, timing, power,
area, and routability tables

Result: the same dataset can support GNNs, image/grid models, classic ML, and exploratory data analysis.

AI/ML Schema OpenStandards Workgroup - Ontology



Original Problem Statement

THE PROBLEM

No standard representation

for AI/ML algorithms and data analytics in EDA.

- Lack of standard makes result comparison and algorithm sharing difficult
- EDA vendor AI/ML investments remain largely closed and proprietary

OPPORTUNITIES

- Industry / academia collaboration
- Reuse of schema specification across organizations
- Transfer ML training without exposing proprietary data

MEMBERS

EDA-Schema · Drexel University (Prof. Ioannis Savidis)
CircuitOps · Nvidia & ASU (Prof. Vidya Chhabria)

AMD · Ansys Synopsys · IBM · Keysight · NXP
Qualcomm · Siemens · Silvaco · Synopsys



Why Ontology, Not Schema

Original Vision
Standardized Schema



EDA Ontology
(RDF / OWL)

What the ontology provides:

Formal Description

Cells, pins, nets, design phases, design activities — formalized as machine-readable entities

Semantic Bridge

Links graph representations, ML datasets, and tool semantics through RDF/OWL formalism

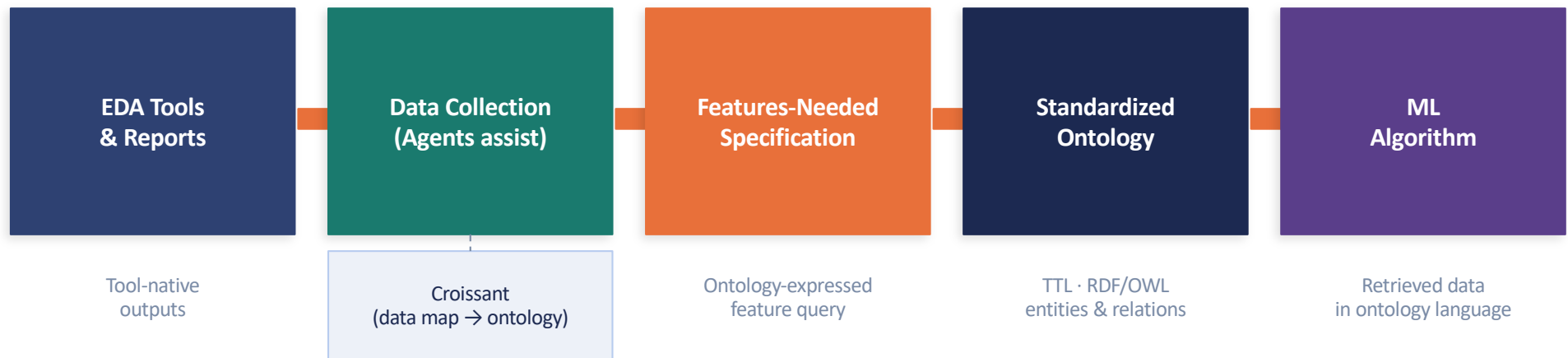
Common Ground

Shared knowledge layer supporting Drexel, ASU use-cases and future collaborative teams

New opportunities beyond the original vision:

- Enables LLM-driven reasoning over EDA domain knowledge
- Supports vendor interoperability for agentic flows
- Enables AI-based EDA flows previously too complex to assemble

System Architecture: Ontology as Semantic Membrane



The Semantic Gap This Closes

Without ontology: agents operate on syntax — tool-specific parsers, translation for every integration

With ontology: entities carry machine-readable meaning; agents discover, query, and reason without human mediation

Pipeline Components to Connect EDA to ML

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Ontology (TTL)

- Defines relationships between entities
- Captures flow-related variables & ML-specific attributes with standard naming conventions

Croissant (JSON)

- MLCommons proposed standard file format for describing datasets

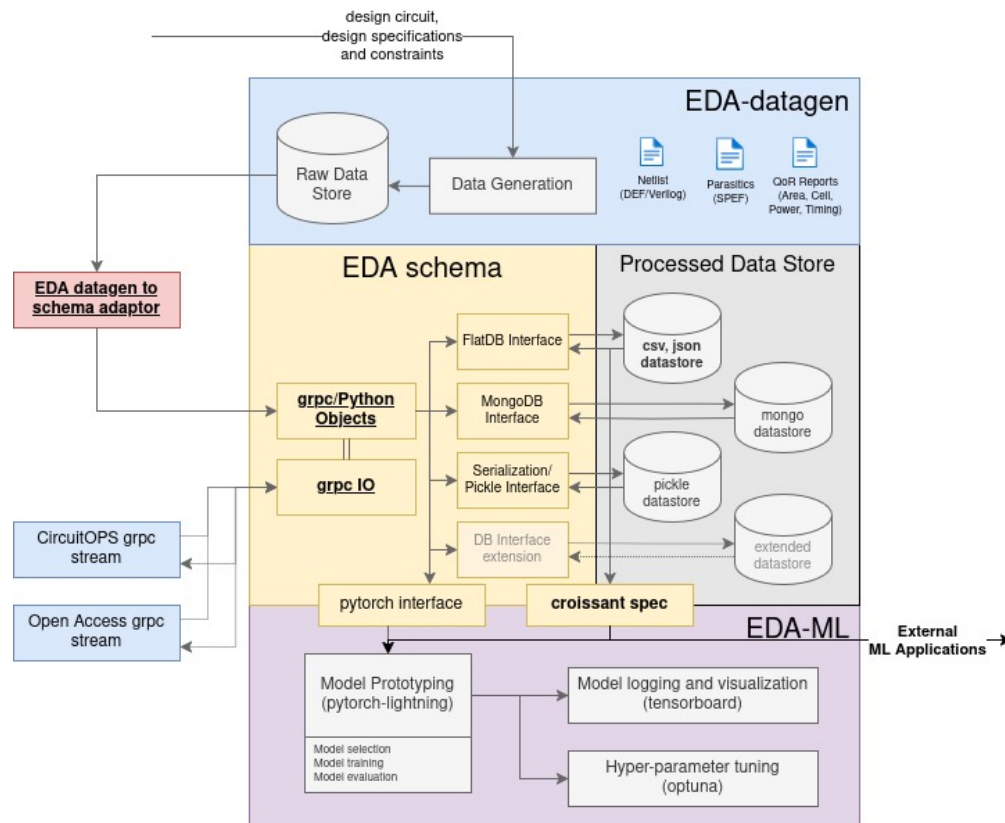
Data source (pandas-df, csv)

- Set of files stored in a structured format containing actual design data

Communication layer (protobuf)

- File to facilitate the transfer of data across servers

Drexel-ICE Labs: Contribution



Support for gRPC

Align the EDA schema with

- taxonomy.json
- ontology.ttl

Regenerate and Re-profile Dataset

- Use OpenROAD + sky130hd PDK with defined constraints
- Re-generate the EDA schema open dataset using the Updated ontology
- Profile Dataset

Retrain and Update ML Models

- Retrain models
- Croissant File regeneration utility

Public Release of AI/ML for EDA Ontology

- Motivation for public release
 - Gather feedback from wide audience
 - Attract additional collaborators
 - Seed a broadly useful open standard
- Contents (approved for public release by Si2 on 3/19/26)
 - EDA ontology (TTL/OWL)
 - Documented, validated use-cases (ASU & Drexel)
 - Ontology validation collateral
 - MCP server – agent discovery & reasoning
 - Documentation package of ontology and use cases
- Released on Github with Apache 2.0 license:
https://github.com/si2org/aiml_schema_ontology
- Si2 AI/ML Schema OWG members retain editorial control



Thank You!

Check out our open dataset at
github.com/drexel-ice/EDA-schema

Check out the EDA ontology at
https://github.com/si2org/aiml_schema_ontology